

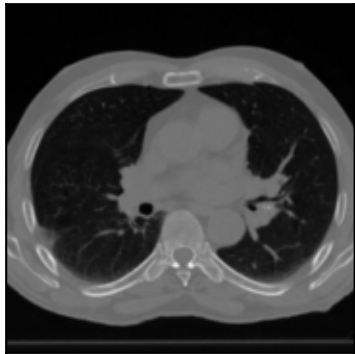
DVF QA pipeline report

DIR-Lab-case3 [intensity_roi]

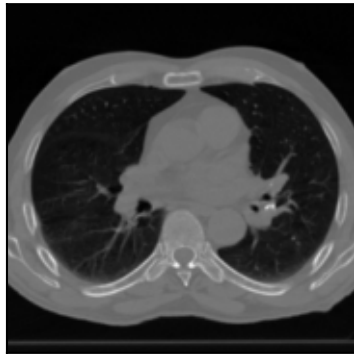
Field	Value
Case	DIR-Lab-case3 [intensity_roi]
Generated	2026-05-20 08:02:36
source	Case3Pack
lateral_strategy	intensity_roi
fps	15.0
n_cycles	4
frames_per_cycle	30
n_phases	10
ct_shape	(104, 256, 256)
spacing_xyz_mm	(1.15, 1.15, 2.5)
detector_shape	(256, 256)
pixel_spacing_mm	(1.0, 1.0)

4DCT phase preview

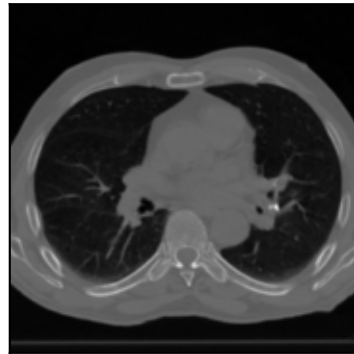
ϕ_0



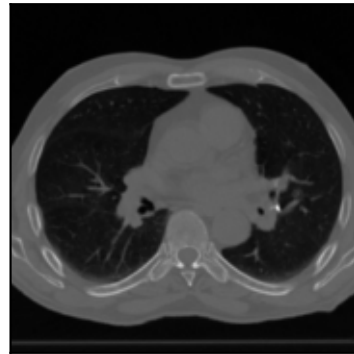
ϕ_1



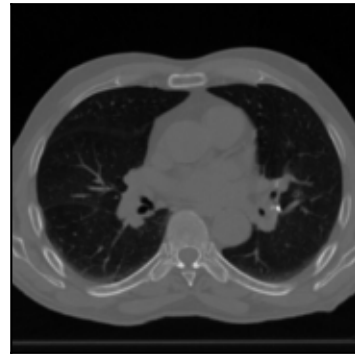
ϕ_2



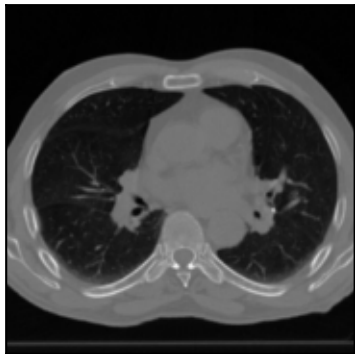
ϕ_3



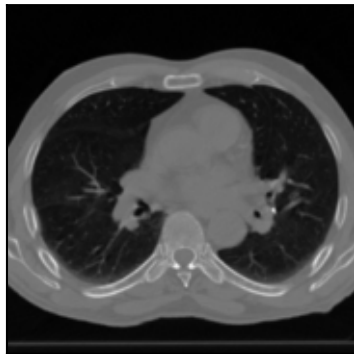
ϕ_4



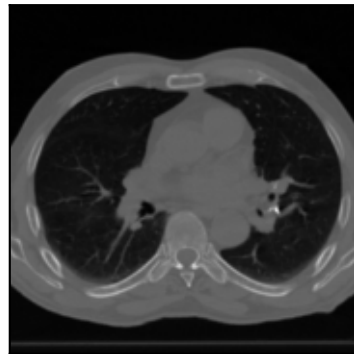
ϕ_5



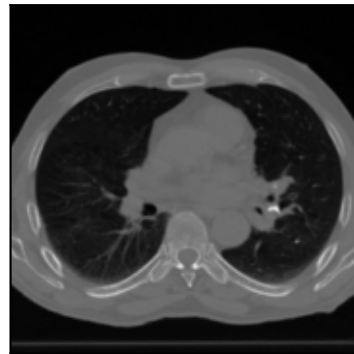
ϕ_6



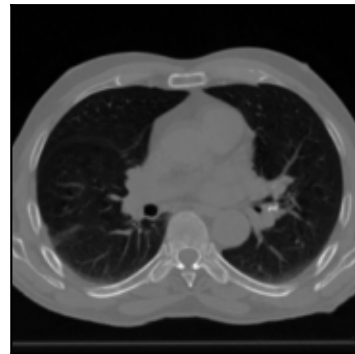
ϕ_7



ϕ_8



ϕ_9



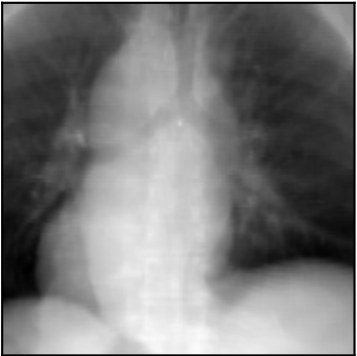
Per-phase DRRs

AP φ_0 

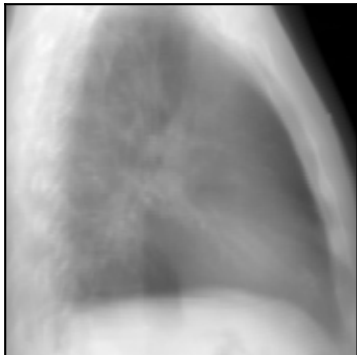
AP $\phi 1$

AP ϕ_2 AP $\phi 3$ AP ϕ_4 AP $\phi 5$ 

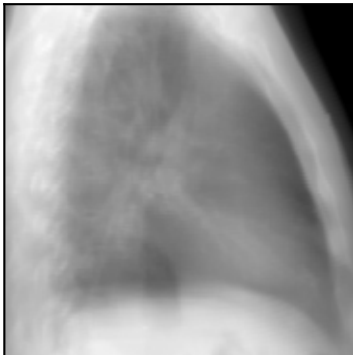
AP φ6

AP $\phi 7$ AP $\phi 8$ 

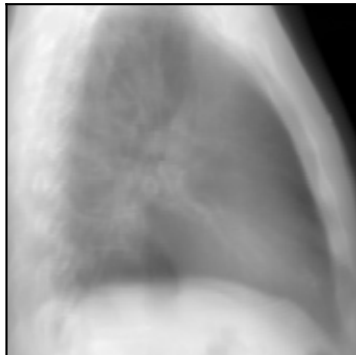
AP φ9

Lat φ_0 

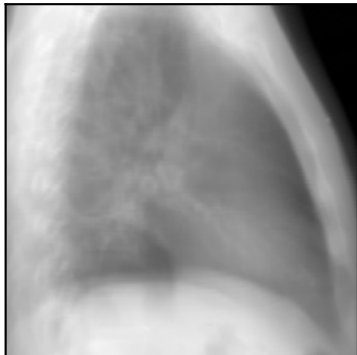
Lat ϕ_1



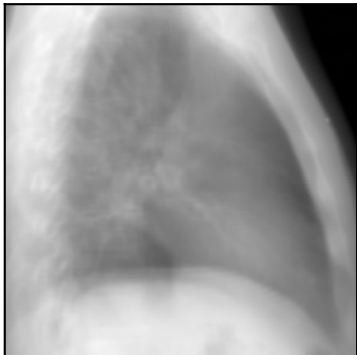
Lat ϕ_2



Lat ϕ_3



Lat ϕ_4



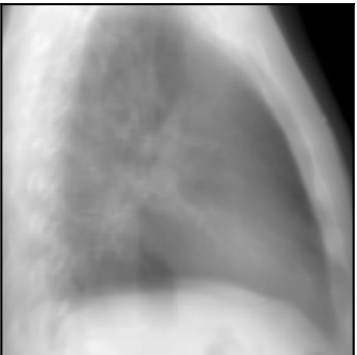
Lat ϕ_5

Lat ϕ_6 

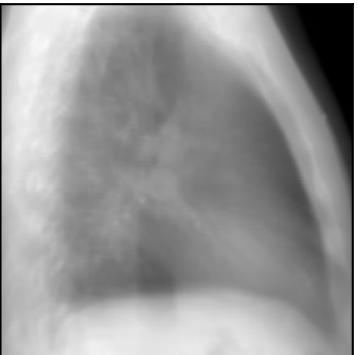
Lat ϕ_7



Lat ϕ_8



Lat ϕ_9



Dynamic sequence samples

AP t=0 (0.00s)



AP t=24 (1.60s)



AP t=48 (3.20s)



AP t=72 (4.80s)



AP t=96 (6.40s)



AP t=120 (8.00s)



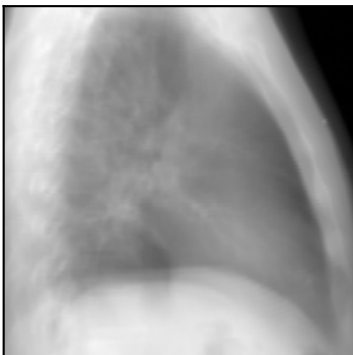
Lat t=0



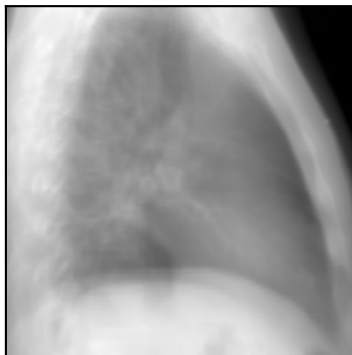
Lat t=24



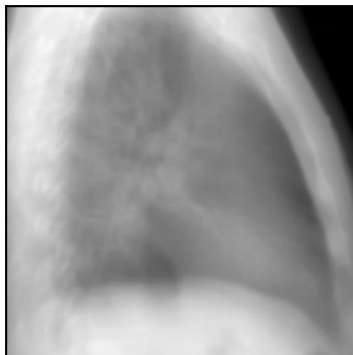
Lat t=48



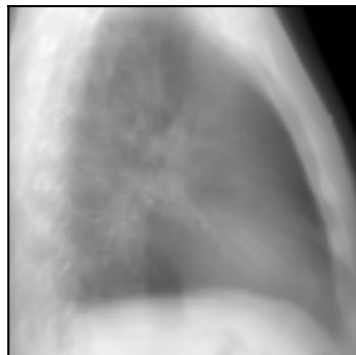
Lat t=72



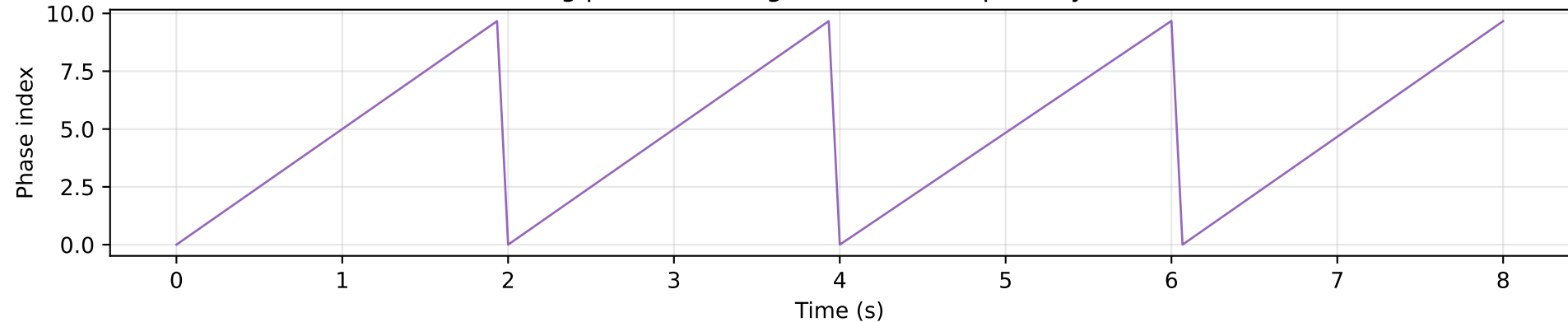
Lat t=96



Lat t=120

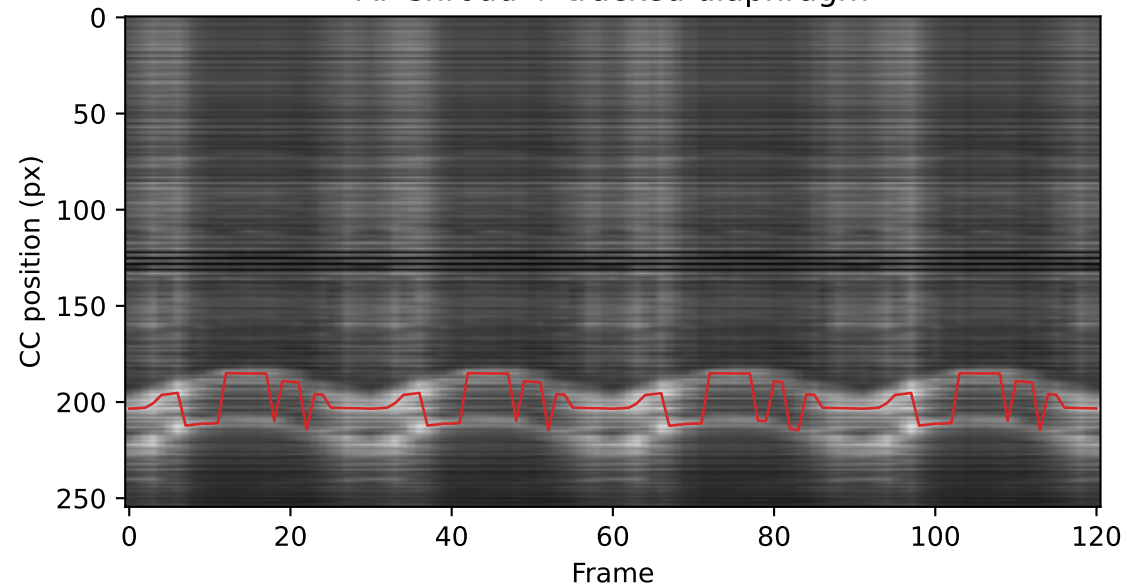


Driving phase curve
Driving phase curve (ground-truth respiratory motion)

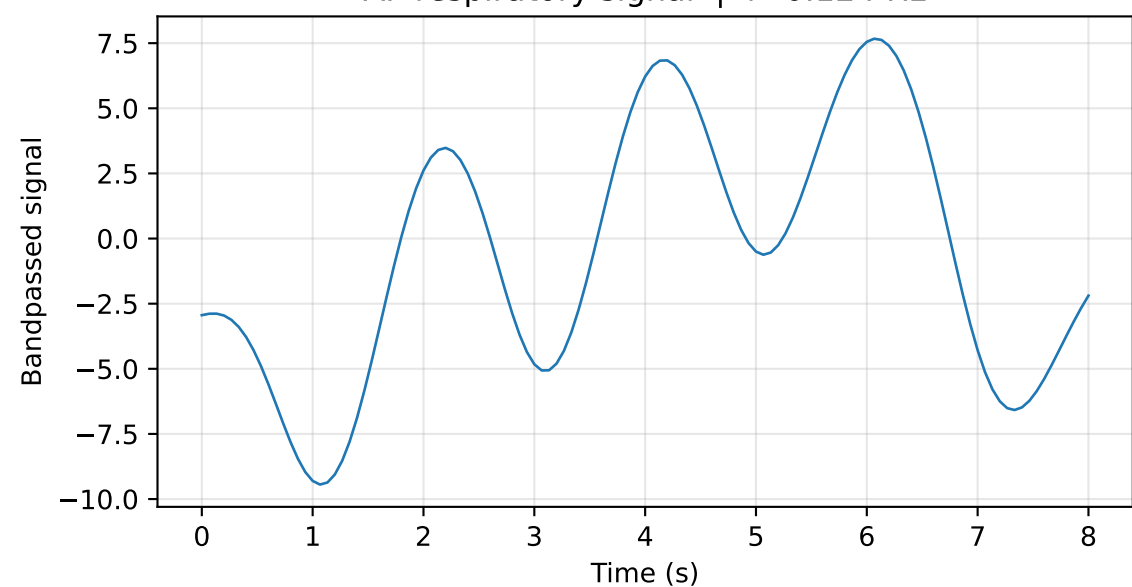


Amsterdam Shroud — AP

AP shroud + tracked diaphragm

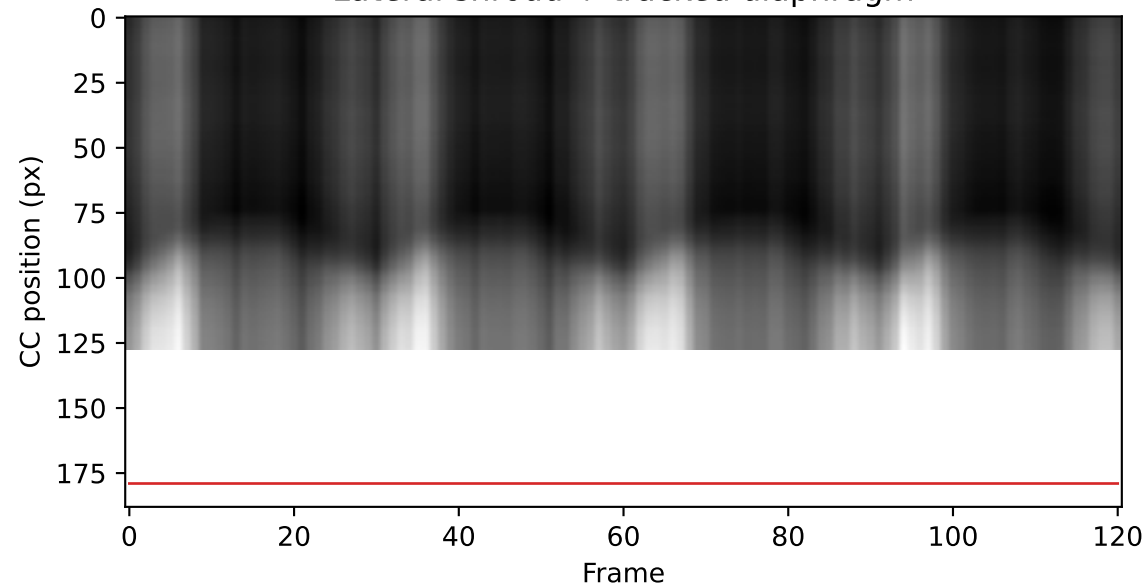


AP respiratory signal | $f=0.124$ Hz

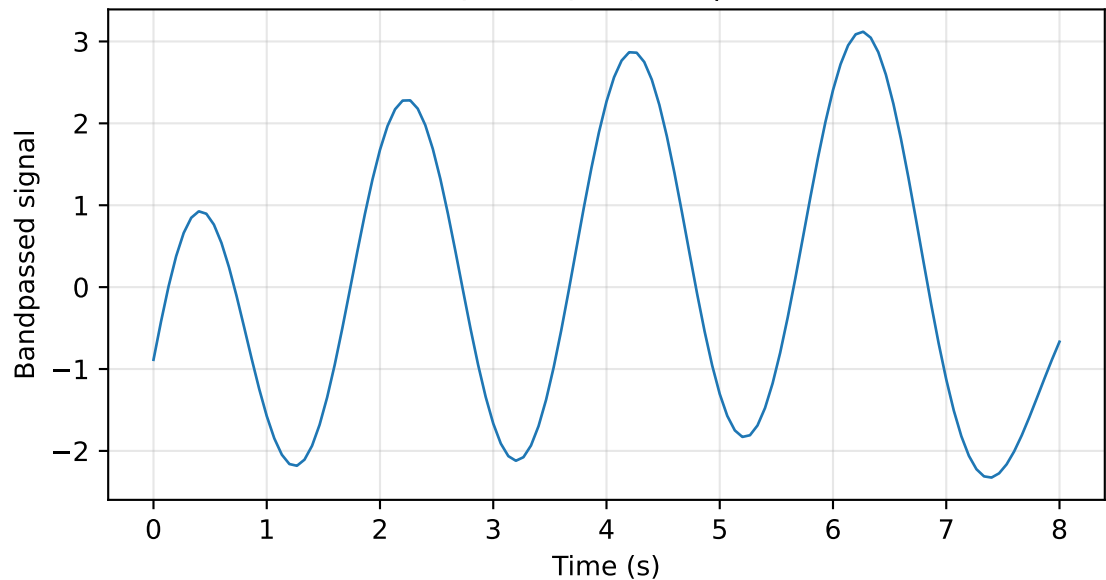


Amsterdam Shroud — Lateral

Lateral shroud + tracked diaphragm

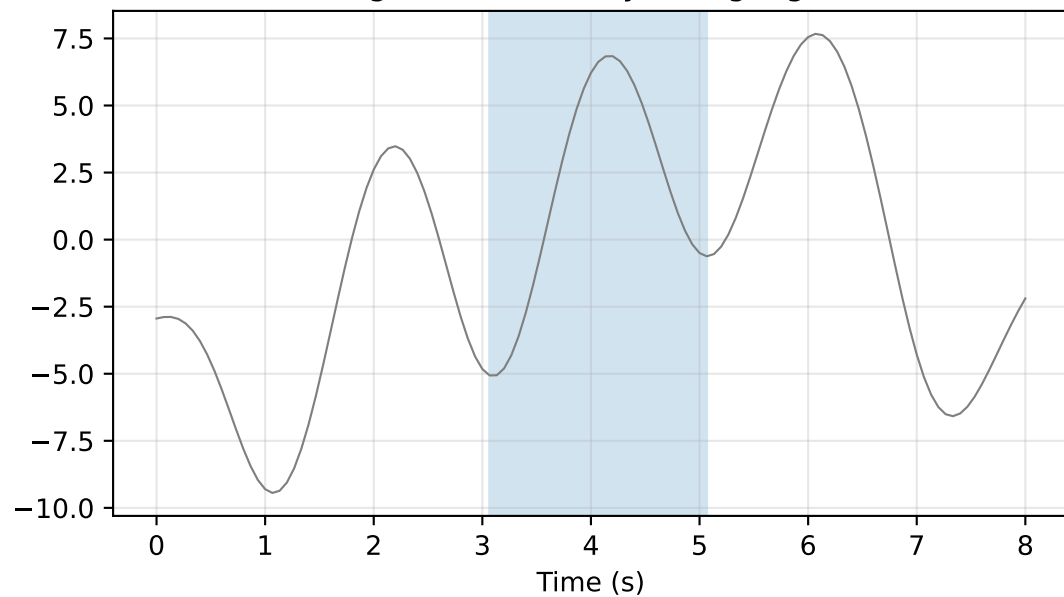


Lateral respiratory signal | $f=0.496$ Hz

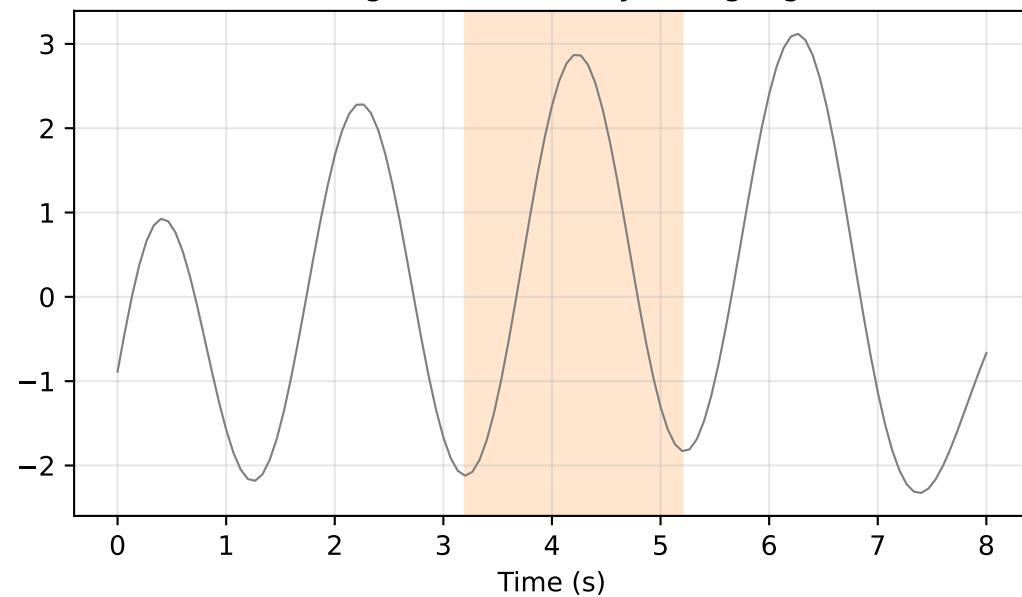


Cycle pairing

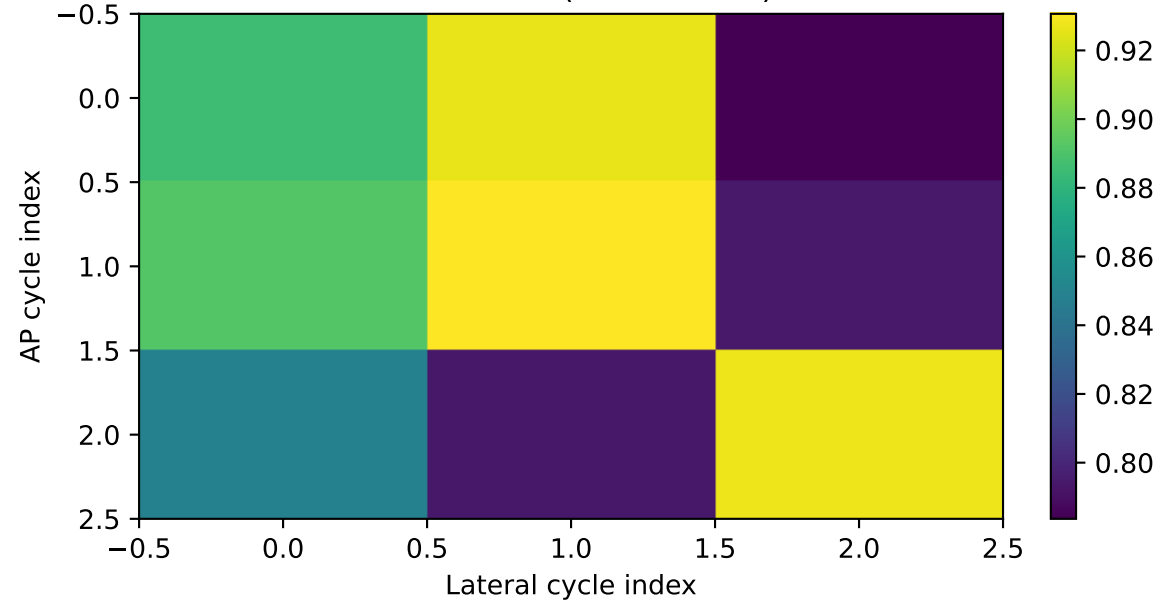
AP signal (selected cycle highlighted)



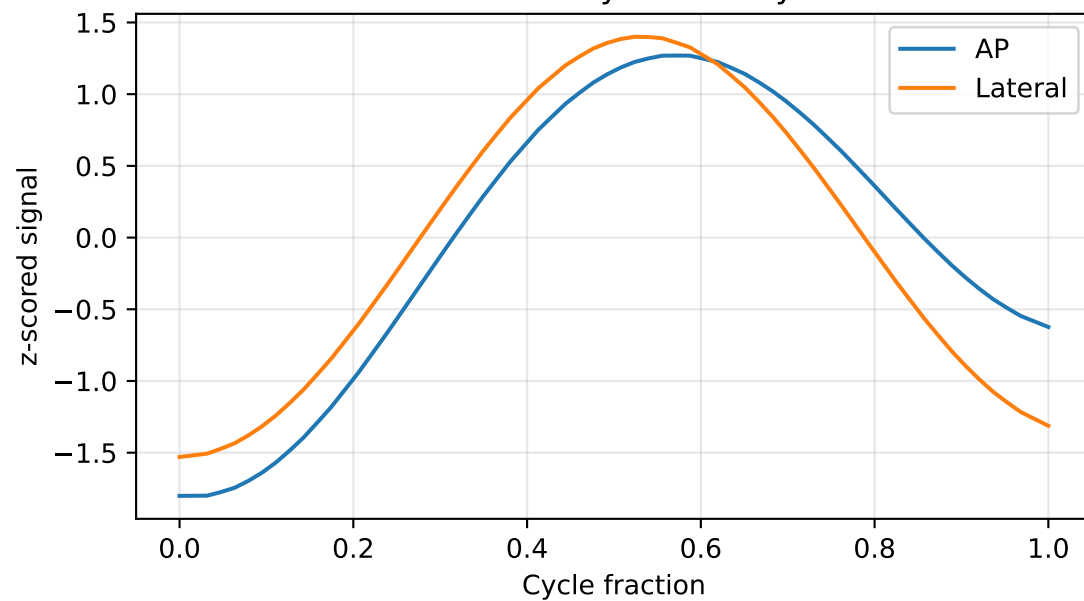
Lateral signal (selected cycle highlighted)



NCC matrix (best=0.931)



Selected cycle overlay



Metrics

Metric	Value
AP dominant frequency (Hz)	0.1240
AP period (frames)	121.00
Lateral dominant frequency (Hz)	0.4959
Lateral period (frames)	30.25
Best pair NCC	0.9306
AP cycle frames	46 - 76
Lateral cycle frames	48 - 78